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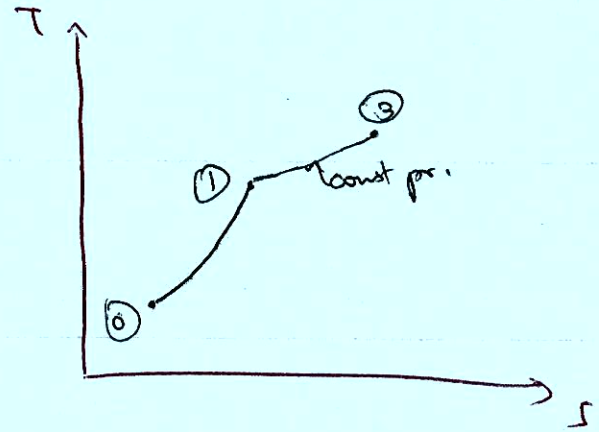

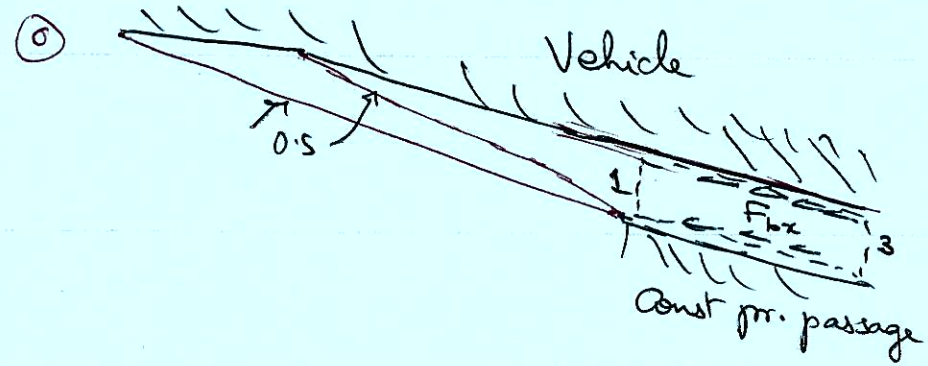

What about frictional losses?

Let F_{xb} denote the total force exerted on the walls due to friction (isobaric flow)

Define skin friction coeff C_f

$$C_f \equiv \frac{F_{xb}}{\frac{1}{2}\rho V^2 A_w} \quad ; \qquad A_w \equiv \text{wall surface area}$$

Can separate our non-isentropic behavior due to shocks & friction

Let $\eta_c' \equiv$ overall isentropic efficiency

$\eta_1' \equiv$ isentropic efficiency of shock structure.

Thus, $\eta_c' < \eta_1'$

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Define an area ratio, $\frac{A_w}{A_3}$ where

$A_w \equiv$ wall surface area.

$A_3 \equiv$ flow thru area @ end of compressor/inlet.

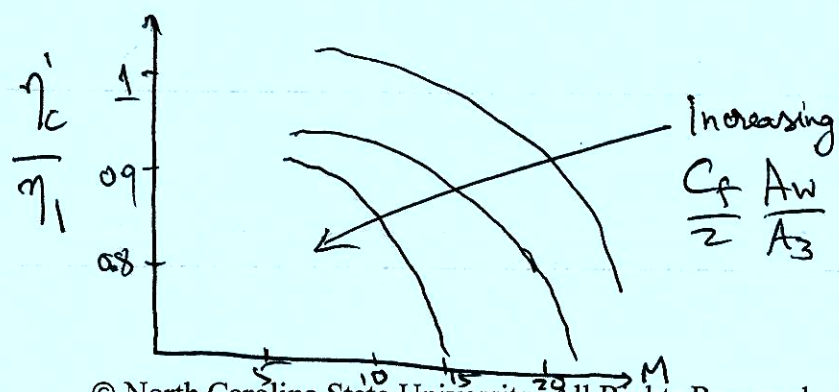
As $M\uparrow$ $\underbrace{C_f\downarrow}_{\text{b/c } Re\downarrow}$ $\frac{A_w}{A_3}\uparrow$ } squeeze the flow more w/ increasing Mach #

B.L. $\delta_0\uparrow \Rightarrow C_f\downarrow$

C_f : ~~0.002~~ 0.002 - 0.005 (encompasses different geometries & Mach #)

$\frac{C_f}{2}\frac{A_w}{A_3}$: $0.01 < \frac{C_f}{2}\frac{A_w}{A_3} < 0.05$ (encompasses different geometries & M: 5 - 20)

Note: C_f is relatively insensitive to M#, but $F_{xb} \sim C_f \times \rho V^2 \sim M_\#^2$ is very sensitive to Mach #.



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- Also note that shock was reflecting off a surface w/ a b.l. & this can cause the b.l. to separate.
- Note even though the core flow is supersonic, the region within the b.l. next to the wall is always subsonic.
- Since information can propage only @ the speed of sound, the only channel thru which u/s propagation of a back pr. fluctuation can happen in a canonical 2D inlet is thru the subsonic region within the b.l. (Note: back pr is the static pr. @ the exit of the compression system which is always higher that the inlet pr.)
- If the back pr. is too high, the inlet can <u>unstart</u> $\Rightarrow$ disgorging of the shock system. This leads to a strong normal shock — η_c goes <u>down</u> $\Rightarrow$ very bad for scramjets.

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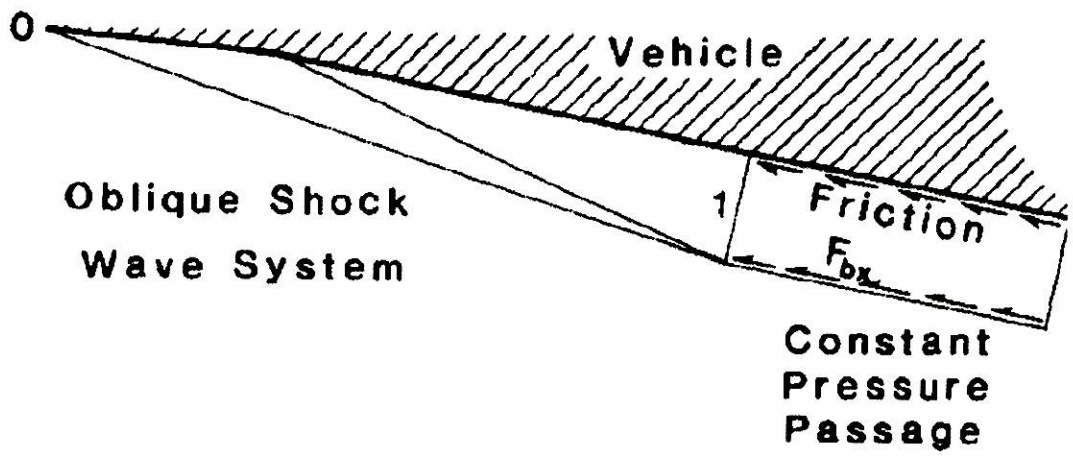

Conceptual analytical model of compression
to estimate the influence of boundary layer
compression efficiency.

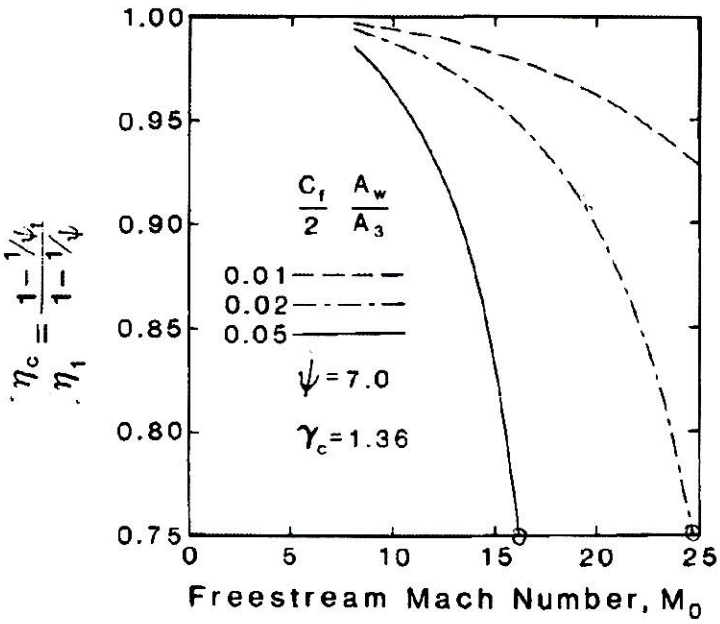

Fig. 5.19 Contribution of boundary-layer friction to adiabatic compression efficiency, from Eq. (5-20), as a function of freestream Mach number and dimensionless skin friction coefficient quantity.

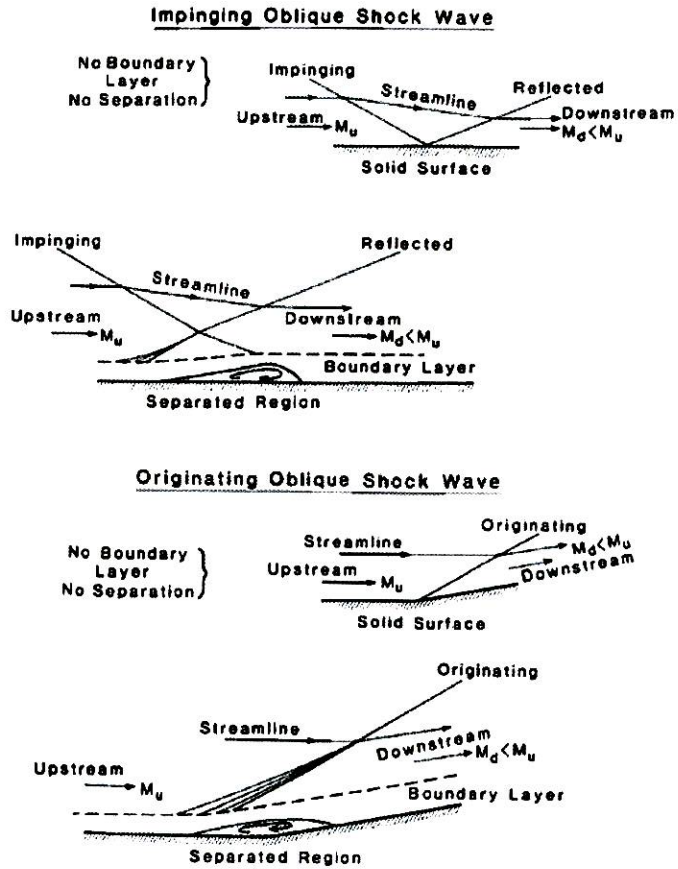

Fig. 5.28 Schematic illustration of boundary-layer separation and reattachment caused by impinging or originating oblique shock waves. Note that the separation or recirculation zone appears to the distant flowfield as a minor perturbation to the geometry of the solid boundary. The downstream compression and expansion waves must therefore eventually coalesce into an oblique shock wave very much like the one expected from the unperturbed boundary.